

KOWANYAMA, Australia

WMO: 942680

Lat: 15.4817S

Lon: 141.7483E

Elev: 11

StdP: 101.20

Time Zone: 10.00 (E10)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	14.3	0.2	3.8	23.0	3.1	4.7	23.5	7.3	26.7	6.6	27.0	0.9	130	0.355

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	10.9	36.8	22.6	36.0	22.7	35.1	23.0	29.4	31.3	27.6	30.6	26.9	30.3	4.2	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.0	25.6	30.4	26.9	22.6	28.8	26.2	21.6	28.2	96.9	31.1	88.0	30.7	85.0	30.4	36.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.0	6.3	5.7	9.5	38.9	2.0	0.9	8.0	39.6	6.9	40.1	5.7	40.6	4.3	41.3
			6.8	33.1	1.4	3.2	5.8	35.4	5.0	37.3	4.2	39.2	3.3	41.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.0	28.5	28.4	28.2	27.4	25.4	23.6	23.1	24.1	26.8	29.1	30.0	29.8
	DBStd	2.84	1.43	1.30	1.16	1.22	1.92	2.14	2.14	1.86	1.67	1.35	1.27	1.52
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	0	0	0	0	0	1	1	0	0	0	0	0
	CDD10.0	6210	574	514	564	522	478	407	407	436	504	591	601	612
	CDD18.3	3169	316	281	306	272	219	157	149	177	254	333	351	354
	CDH23.3	33761	3422	2969	3156	2753	2081	1398	1443	1744	2554	3704	4256	4281
(13)	CDH26.7	15016	1352	1120	1241	1195	876	512	528	749	1308	1899	2159	2076
(14)	Wind	WSAvg	2.6	2.5	2.4	2.3	2.3	2.5	2.6	2.7	2.9	3.1	3.0	2.7
(15) Precipitation	PrecAvg	1297	322	392	233	54	4	2	2	1	2	19	64	167
	PrecMax	2000	790	848	686	261	68	24	14	15	33	284	254	437
	PrecMin	565	60	42	23	0	0	0	0	0	0	0	0	18
	PrecStd	362	177	176	147	60	11	5	3	4	7	46	62	97
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.5	34.7	34.5	34.4	34.0	32.5	32.7	34.1	36.3	37.7	37.7	37.9
		MCWB	24.6	26.0	25.5	24.2	22.6	21.3	20.9	21.3	21.0	21.6	21.9	22.2
	2%	DB	34.1	33.7	33.5	33.5	32.9	31.5	31.6	32.8	35.1	36.5	36.7	36.4
		MCWB	25.4	26.1	25.3	23.9	22.3	21.1	20.5	20.2	20.5	21.8	22.6	23.9
	5%	DB	33.2	32.9	32.8	32.7	32.0	30.7	30.7	31.7	34.2	35.5	35.8	35.3
		MCWB	25.7	26.0	25.2	23.7	22.3	20.7	19.9	19.4	20.3	21.9	23.1	24.2
	10%	DB	32.3	32.0	32.0	32.1	31.1	29.8	29.8	30.7	33.1	34.6	34.9	34.2
		MCWB	25.7	26.1	25.1	23.5	21.9	20.2	19.4	18.9	20.2	21.9	23.5	24.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.6	29.1	28.5	31.4	30.4	25.2	25.8	27.0	27.2	30.4	31.8	30.1
		MCDB	30.8	32.7	31.3	31.5	31.0	26.5	29.1	28.3	30.3	31.0	34.0	31.2
	2%	WB	27.3	27.9	27.2	26.4	25.8	22.7	22.7	22.2	23.8	25.5	27.1	27.1
		MCDB	30.7	31.0	30.5	29.8	30.0	28.5	28.3	28.9	29.1	29.2	31.1	31.1
	5%	WB	26.8	27.2	26.7	25.6	24.2	22.0	21.6	21.2	22.9	24.7	26.1	26.6
		MCDB	30.2	30.2	29.9	29.2	27.9	28.2	27.7	27.7	28.4	29.0	30.3	30.5
	10%	WB	26.4	26.7	26.2	25.0	23.4	21.4	20.7	20.5	22.2	24.1	25.5	26.2
		MCDB	29.7	29.7	29.4	28.8	27.5	27.4	26.8	26.8	27.7	28.9	29.9	30.0
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	7.1	7.9	10.1	12.2	13.0	14.1	14.9	14.5	12.8	10.9	9.3
		MCDBR	9.0	8.6	9.0	10.7	12.4	13.2	13.9	15.2	14.9	13.5	11.9	11.1
		MCWBR	3.0	2.9	2.8	3.4	4.3	4.9	5.0	5.3	4.4	4.5	4.4	3.7
	5% WB	MCDBR	7.6	7.4	7.9	9.0	10.4	11.9	12.2	13.9	13.5	11.9	10.3	8.8
		MCWBR	3.1	3.1	3.1	4.1	5.4	5.2	6.0	5.9	4.8	4.9	4.9	3.5
(40) Clear-Sky Solar Irradiance	taub		0.410	0.404	0.387	0.370	0.356	0.344	0.338	0.349	0.394	0.424	0.448	0.437
	taud		2.509	2.529	2.574	2.597	2.584	2.592	2.594	2.546	2.407	2.343	2.310	2.382
	Ebn at Noon		935	932	927	910	889	884	899	917	906	903	893	909
	Edh at Noon		115	111	103	95	90	87	88	98	120	132	139	130
(44) All-Sky Solar Radiation	RadAvg		5.77	5.73	5.86	5.85	5.19	4.75	5.07	5.79	6.68	7.23	7.23	6.53
	RadStd		0.55	0.74	0.51	0.37	0.23	0.25	0.16	0.14	0.15	0.25	0.39	0.69

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page